

System Analysis

System Request & Feasibility Study

Project

Smart Queue Management System

Planning Phase

Course	System Analysis
Project	Smart Queue Management System
Study	System Request and Feasibility Study

System Request

The System Request describes the reasons for building the system and the value that the system is expected to provide. It contains the project sponsor, business need, business requirements, business value, and special issues.

Project Sponsor

Customer Service Manager

The Customer Service Manager is the person who initiates and supports the project.

Business Need

- Reduce customer waiting time.
- Reduce overcrowding in service areas.
- Improve customer service.
- Organize the customer queue.
- Give customers better information about their waiting time.

Business Requirements

- Allow customers to take a queue number.
- Display the current queue number.
- Allow employees to call the next customer.
- Provide an estimated waiting time.
- Notify customers when their turn is approaching.
- Generate reports about queue performance.
- Store information about completed services.

Business Value

Expected Benefit	Estimated Value
Reduction in average waiting time	30% reduction
Increase in customers served per day	20% increase
Reduction in customer complaints	15% reduction
Annual operational cost savings	120,000 EGP
Customer service improvement	Higher customer satisfaction

Special Issues

- The system should be completed within 4 months.
- Customer data must be protected.

- The system should be easy for employees to use.
- The system should support at least 500 customers per day.
- The system should work with the organization's existing computers.

Feasibility Study

The feasibility study helps determine whether the project can be developed and used successfully. It includes technical, economic, and organizational feasibility.

Technical Feasibility

Can We Build It?

Factor	Evaluation
Familiarity with application	High - The team understands the basic requirements of queue management systems.
Familiarity with technology	High - The required technologies are commonly used and available.
Project size	Medium - The project can be developed by a small development team.
Compatibility	High - The system can work with the existing computers and network.

Conclusion: The project is technically feasible because the required technology is available and the development team can build the system.

Economic Feasibility

Should We Build It?

Development Costs

Item	Cost (EGP)
Software Development	70,000
Hardware	45,000
Database & Server Setup	20,000
Testing & Deployment	15,000
Total Development Cost	150,000

Annual Operating Costs

Item	Annual Cost (EGP)
Server & Hosting	18,000
Maintenance	20,000
Technical Support	12,000
Total Annual Operating Cost	50,000

Annual Benefits

Benefit	Estimated Annual Value (EGP)
Reduced operational costs	120,000
Increased service capacity	80,000
Reduced paperwork	30,000
Total Annual Benefits	230,000

First-Year Summary

Item	Amount (EGP)
Development Cost	150,000
Annual Operating Cost	50,000
Total First-Year Cost	200,000
Total Annual Benefit	230,000
Expected First-Year Net Benefit	30,000

Conclusion: Based on the estimated costs and benefits, the project has an expected first-year net benefit of 30,000 EGP.

Organizational Feasibility

If We Build It, Will They Use It?

Stakeholder	Expected Impact
Senior Management	Management can monitor waiting times and service performance.
Employees	Employees can call and manage customers through the system.
Customers	Customers can receive queue numbers and estimated waiting information.
IT Department	The system can work with the organization's existing environment.

Conclusion: The system is organizationally feasible because the main stakeholders are expected to benefit from using it.

Overall Feasibility

Area	Result
Technical Feasibility	Feasible
Economic Feasibility	Feasible
Organizational Feasibility	Feasible

Overall Project Conclusion

The Smart Queue Management System appears feasible based on the technical, economic, and organizational analysis. The system can be developed using available technology, provides measurable expected benefits, and addresses the needs of management, employees, and customers.